Financial Modelling Assignment

Words
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1. Introduction:

Creating and modeling portfolios, is vital for analysts seeking to maximize returns and manage risk. By leveraging these models, I can guide investment strategies, enhance portfolio performance, and provide actionable recommendations tailored to varying market conditions and client objectives.

This report presents my perspective of a Fund Analyst specializing in US Healthcare stocks, the Fund Manager has asked me to create an Optimal Active Portfolio from a set of US based Healthcare company stocks, furthermore, this report provides benchmarks for the Tangency Portfolio and Market Portfolio in correlation with the optimal active portfolio.

The report illustrates the techniques for modelling portfolios, which employ quantitative approaches to portfolio optimization, highlight the intuition and objective behind modelling portfolios.

Data: This section gives a brief about the input data and provides summary statistics for the assets.

VaR & CVaR: Introduces the calculation of Value at Risk (VaR) and Conditional Value at Risk (CVaR) using the historical simulation approach. By analyzing past market data, this method estimates potential future losses, providing crucial risk management insights. VaR quantifies the maximum expected loss, while CVaR measures the average loss beyond the VaR threshold.

Sensitivity Analysis: Individual analyst's expectations vs corresponding portfolio weights analysis, Mean returns and Sharpe ratio analysis and Variance Co-variance VaR vs Historical Simulation VaR Analysis were performed to gain more insights into understand the optimal active portfolio.

Finally, a comprehension of the reasons behind the distinctiveness and variations of each portfolio, in addition to the major findings and critical evaluations of shortcomings, as well as recommendations for enhancing certain techniques to produce a better-modelled portfolio.

2. Methodology:

In this analysis, we explore portfolio optimization within the mean-standard deviation space, a two-dimensional framework where mean returns are plotted on the x-axis and standard deviation values on the y-axis. I have constructed a portfolio that has the least risk, as no specific constraints on risk or expected return preferences are provided by the investor.

Our methodology leverages the Markowitz mean-variance model, which identifies an efficient frontier—a curve representing portfolios that offer the highest expected return for a given level of risk. This efficient frontier is crucial in guiding the selection of optimal portfolios.

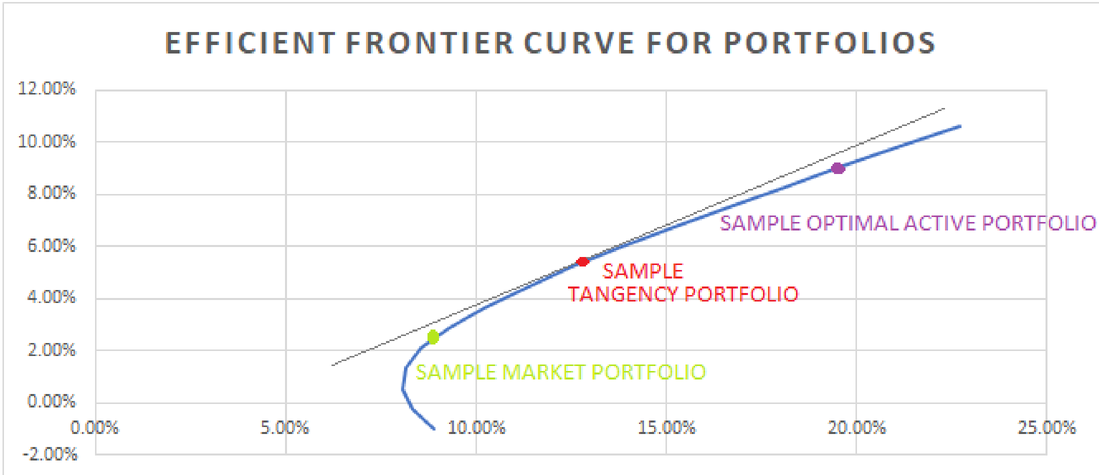


Figure.1: Efficient Frontier curve graph.

Within the parabola encapsulating the efficient frontier curve, we identify the tangency portfolio, the market portfolio (generally representing a broad market index), and the optimal active portfolio (designed to enhance returns through active management). Mapping these portfolios in the mean-standard deviation space provides a clear visualization of their risk-return profiles, thereby aiding investors in making informed decisions.

It is important to note that portfolios can lie anywhere in the Cartesian plane. However, it is most suitable for an investor to focus on portfolios located within the parabola or on the efficient frontier line. Portfolios outside these points may entail excessive risk, insufficient returns, or potential losses, making them less desirable for investment.

2.1. Tangency Portfolio:

The Tangency Portfolio weights for the eight US based Healthcare stocks can be calculated by the below formula:

$$[Tangency\ Portfolio]_{8*1} = \frac{[Covariance\ Matrix]_{8X8}^{-1} * ([Mean\ Returns]_{1X8}^{T} - r_f)}{\sum_{i=1}^{i=8} \sum_{j=1}^{j=8} ([Covariance\ Matrix]_{8X8}^{-1} * \left([Mean\ Returns]_{1X8}^{T} - r_f\right))} \tag{1}$$

Where,

$$[Covariance\ Matrix]_{8X8}^{-1} = Inverse\ of\ the\ Covariance\ matrix\ of\ the\ 8\ asset\ portfolio,$$

$$[Mean\ Returns]_{1X8}^{T} = Transpose\ of\ the\ Mean\ returns\ matrix,$$

$$r_f = risk\ free\ rate,$$

The numerator represents an 8X1 matrix obtained after multiplication of inverse of covariance matrix and mean returns matrix minus the risk-free rate, the denominator represents the sum of all values in matrix obtained in the numerator.

2.1.1. Sharpe Ratio Maximization of Tangency Portfolio:

The Sharpe ratio evaluates the relationship between an investment's risk and return. It highlights that excess profits over time may reflect increased risk and volatility rather than the ability in investing.

Sharpe ratio formula:

$$Sharpe\ ratio = \frac{[Mean\ return]_{1X8} * [Tangency\ portfolio]_{8X1} - r_f}{\sqrt{[Tangency\ portfolio]^T_{8X1} * [Covariance\ matrix]_{8X1} * [Tangency\ portfolio]_{8X1})}} \quad (2)$$

2.2. Market Portfolio:

- The Capital Asset Pricing Model (CAPM) provides a framework for estimating the expected return of an asset based on its risk relative to the market.
- From the above context, the market portfolio is used as a benchmark, with its weights determined by dividing the market capitalization of each asset by the total market capitalization of all assets in the portfolio.

$$[Market\ Portfolio]_{8*1} = \frac{[Market\ Capitalisation_i]_{i=1}^{8}}{\sum_{i=1}^{i=8} (Market\ capitalization_i)} \quad (3)$$

2.3. Optimal Active Portfolio:

The optimal active portfolio is determined using the Black-Litterman approach, which combines the market's perspective on returns with the analyst's views on expected return rates. This dual-process method enhances the precision of portfolio allocation by integrating both market data and analyst insights.

The approach is detailed as follows below.

2.3.1. Incorporating markets perspective of expected returns:

The market implied expected returns are formulated as below:

$$[Market\ implied\ ER]_{8X1} = \omega * [Covariance\ matrix]_{8X8} * [Market\ Portfolio]_{8X1} + r_f \quad (4)$$

$$Where, \quad \omega = \frac{Market\ Expected\ Return - r_f}{Market\ Variance} \quad (5)$$

2.3.2. Incorporating the Analyst's perspective of their views on the assets expected return into the portfolio:

The analyst's perspectives can be integrated by calculating both the Analyst's Expected Return and the Expected Return adjusted for these views. These views can be documented in a single-column table, where the percentage changes in expected returns for each asset in the portfolio are manually entered based on the analyst's assumptions. Assets without specific return assumptions can be recorded with a zero percent change.

$$[ER\ Adj\ for\ 2\ views]_{8X1} = [Market\ Exp\ Ret]_{8X1} * [Tracking\ Factor\ Matrix]_{8X8} * [Adj\ Delta]_{8X1} \quad (6)$$

The Tracking Factor matrix represents an asset's sensitivity relative to another stock, accounting for correlations between various portfolio assets. It is calculated by dividing the covariance between two assets by the variance of one of the assets. This matrix is crucial for understanding inter-asset relationships within the portfolio.

Finally, the Optimal Active Portfolio weights are constructed using the below formula:

$$[Opt\ Active\ Portfolio]_{8X1} = \frac{[Covariance\ Matrix]_{8X8}^{-1} * \left([Expected\ Return\ adjusted\ for\ views]_{8X1} - r_f\right)}{\sum_{j=1}^{8}\sum_{i=1}^{8}[Covariance\ Matrix]_{8X8}^{-1} * ([Expected\ Return\ adjusted\ for\ views]_{8X1} - r_f)} \quad (7)$$

Where, $\sum_{j=1}^{8}\sum_{i=1}^{8}[Covariance\ Matrix]_{8X8}^{-1} * \left([Expected\ Return\ adjusted\ for\ views]_{8X1} - r_f\right) =$ *single integer obtained by adding all values in the matrix obtained after matrix Multiplication of covariance matrix and expected return matrix.*

2.4. Value at Risk:

- Historical Simulation Value at Risk (VaR) is a risk management method that estimates potential portfolio losses by utilizing historical returns, without relying on a predefined distribution.
- This approach offers a non-parametric way to evaluate the risk of significant losses by examining real historical market scenarios under typical market conditions.

Step 1:

Calculation of standardized returns:

$$R_t^s = \frac{R_t - \mu}{\sigma} \quad (8)$$

Where,

R_t^s = *Standardized return,* R_t = *Portfolio return,* μ=*Mean return of the portfolio and* σ= *Standard deviation of the portfolio.*

Step 2:

Use the distribution of standard returns to calculate VaR:

$$VaR = -(q_\alpha^s \sigma + \mu) \quad (9)$$

Where,

q_α^s = *the α-percent quantile of the empirical distribution of standardized returns.*

To estimate VaR for zero mean, simply equate μ=0 in eqn (9), i.e. $VaR = -(q_\alpha^s \sigma)$ (10)

Step 3:

Computing for the time horizon:

$$VaR = \sqrt{T} * VaR \quad (11)$$

Where, *T= Time period, in our case we calculate 1-month VaR hence this value =1.*

2.5. Conditional Value at Risk:

Conditional Value at Risk (CVaR) is a metric that estimates the expected loss beyond the Value at Risk (VaR), offering a deeper insight into potential extreme losses within a portfolio.

CvaR using standardized returns:

$$Q_t^s = E(R_t^s | R_t^s < q_\alpha^s) \quad (12)$$

$$CVaR = -Q_\alpha^s \sigma \quad [For\ zero\ mean\ condition] \quad (13)$$

3. Data:

3.1. Input Data:

- The data used for this analysis has 61 observations which represent closed adjusted price data over a monthly frequency, this data is obtained from yahoo finance. Furthermore, I am provided with 3-month US treasury bill rate of 5.186%, Annualized Expected market returns of 12.5% and market capitalization of each of the US healthcare stocks in billions of dollars.
- The risk-free rate which is the US treasury bond is altered to 3 month compounded returns which values to about 0.432%.
- The tables utilized for portfolio modeling are based on 5 years of simple returns, spanning from January 1, 2019, to December 1, 2023, with data input at monthly intervals.

Table 1: Market Capitalization table (given) of the US Healthcare stocks.

US Healthcare stocks	Market Capitalization (Billions USD)
Johnson and Johnson (JNJ)	385.4
Eli Lilly (LLY)	599.1
AbbVie (ABBV)	290.1
Merck (MRK)	305.4
Pfizer (PFE)	155.2
Bristol-Myers Squibb (BMY)	101.5
Amgen (AMGN)	166.2
Gilead Sciences (GILD)	99.4
Total Market Capitalization:	**2102.3**

3.2. Summary Statistics Data:

- The most robust and consistent returns are shown by LLY, which has the greatest mean (3.19%) and median (2.95%).

Table 2: Summary statistics derived from the simple returns of US Healthcare stocks over the 5-year period.

US Healthcare stocks	mean	median	min	max	range	variance	skewness	kurtosis
JNJ	0.67%	1.07%	-9.09%	14.42%	23.51%	0.002	0.1330	-0.1738
LLY	3.19%	2.95%	-11.86%	23.18%	35.04%	0.007	0.1396	-0.3592
ABBV	1.52%	0.88%	-11.84%	22.89%	34.73%	0.005	0.4540	0.2735
MRK	1.11%	1.08%	-14.92%	17.51%	32.43%	0.004	0.1792	0.7879
PFE	0.01%	-1.30%	-14.38%	23.92%	38.29%	0.006	0.7705	0.7524
BMY	0.41%	-0.75%	-10.78%	16.26%	27.04%	0.003	0.4292	0.2748
AMGN	1.15%	0.79%	-14.64%	19.94%	34.59%	0.005	0.4251	0.0057
GILD	0.99%	-0.57%	-12.06%	27.18%	39.24%	0.005	1.0592	2.3433

- PFE and BMY have negative median values (-1.30% and -0.75%), indicating frequent losses despite PFE's minimal positive mean (0.01%).
- GILD exhibits the highest range (39.24%) and significant volatility (variance of 0.005).
- PFE and MRK also show high ranges (38.29% and 32.43%), reflecting substantial return variability.
- Variance across all stocks remains low (0.002 to 0.007), suggesting relative stability in returns.
- Most stocks show slight positive skewness, with GILD (1.0592) and PFE (0.7705) indicating returns are more frequently higher than the mean.
- Kurtosis values are generally low, with GILD (2.3433) having the highest, indicating more frequent extreme returns compared to a normal distribution.

LLY stands out for its consistent and high returns, while GILD shows the greatest volatility. PFE and BMY's negative medians suggest frequent losses. Overall, the stocks exhibit low variance and mixed skewness and kurtosis, indicating varied return distributions.

4. Results:

Table 3: Black Litterman portfolio optimization outputs. The Adj delta column is obtained by minimizing an objective function of Expected Returns adjusted for the two views.

Assets	Market Implied ER	Analyst Views	Analyst ER
JNJ	0.90%	0.00%	0.90%
LLY	1.21%	0.00%	1.21%
ABBV	0.87%	0.00%	0.87%
MRK	0.88%	0.20%	1.08%
PFE	0.93%	0.30%	1.23%
BMY	0.87%	0.00%	0.87%
AMGN	1.01%	0.00%	1.01%
GILD	0.78%	0.00%	0.78%
Assets	**Adj delta**	**ER adjusted for two views**	**Active Portfolio**
JNJ	0.00%	1.06%	13.72%
LLY	0.00%	1.35%	21.32%
ABBV	0.00%	1.00%	10.33%
MRK	0.16%	1.08%	23.76%
PFE	0.26%	1.23%	17.81%
BMY	0.00%	0.99%	3.61%
AMGN	0.00%	1.18%	5.92%
GILD	0.00%	0.88%	3.54%

The adjusted delta obtained for Merck and Pfizer is estimated 0.04% lower than the actual estimate of analyst's views to 0.16% and 0.26%. This indicates a bullish sentiment from analysts for these specific assets.

The active portfolio allocation is most heavily weighted towards MRK (23.76%) and LLY (21.32%), reflecting the portfolio's strategic tilt towards these assets.

4.1. Estimated Portfolios:

- The Tangency portfolio exhibits extreme weights, such as LLY being significantly overweighted (186.56%) and BMY and JNJ being underweighted (-141.22% and -96.71%, respectively) [Table 4], compared to their Market portfolio weights. This suggests that the Tangency portfolio is highly optimized for risk-return efficiency but may carry high concentration risk.

- The Optimal Active portfolio moderates the extreme allocations of the Tangency portfolio, aligning closer to the Market weights but still actively overweighting MRK (23.76%) and underweighting JNJ (13.72%) [Table 4]. This approach balances aggressive optimization with practical diversification, aiming for enhanced returns without excessive risk.

Table 4: Below table depicts the portfolio weights of benchmark portfolios along with the active portfolio.

US based Healthcare Stocks	Tangency	Market	Optimal Active
JNJ	-96.71%	18.33%	13.72%
LLY	186.56%	28.50%	21.32%
ABBV	138.03%	13.80%	10.33%
MRK	52.48%	14.53%	23.76%
PFE	-68.54%	7.38%	17.81%
BMY	-141.22%	4.83%	3.61%
AMGN	34.16%	7.91%	5.92%
GILD	-4.75%	4.73%	3.54%
Total	**100.00%**	**100.00%**	**100.00%**

- The Black-Litterman model has generated an optimal portfolio allocation without considering the shorting of assets. Instead, it has adjusted the weights by reducing the allocation to the underweighted assets, as opposed to the more extreme short positions seen in the tangency portfolio.

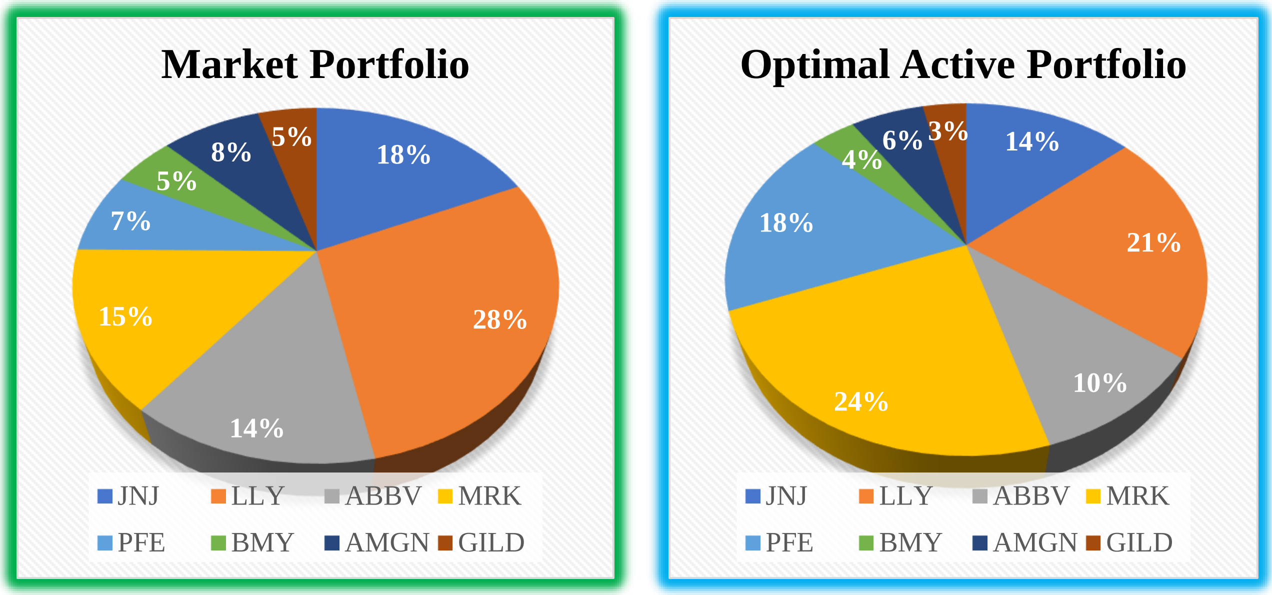


Figure.2: The optimal active portfolio closely resembles the market portfolio in its allocation.

- The Tangency Portfolio offers the highest mean return (7.74%) but with significantly greater risk (16.64% standard deviation), reflecting a more aggressive strategy. Its Sharpe Ratio (0.4392) [Table 4] is superior, indicating better risk-adjusted returns than both the Market and Optimal Active Portfolios.
- Conversely, the Optimal Active Portfolio takes a conservative approach, achieving slightly lower returns (1.31%) with reduced risk (4.62% standard deviation) [Figure.3], aligning closely with market expectations.

Table 4: Mean, Standard deviation and Sharpe ratio for the three portfolios.

	Tangency	Market	Optimal Active
Mean	7.74%	1.56%	1.31%
Std deviation	16.64%	4.70%	4.62%
Sharpe ratio	0.4392	0.2405	0.1907

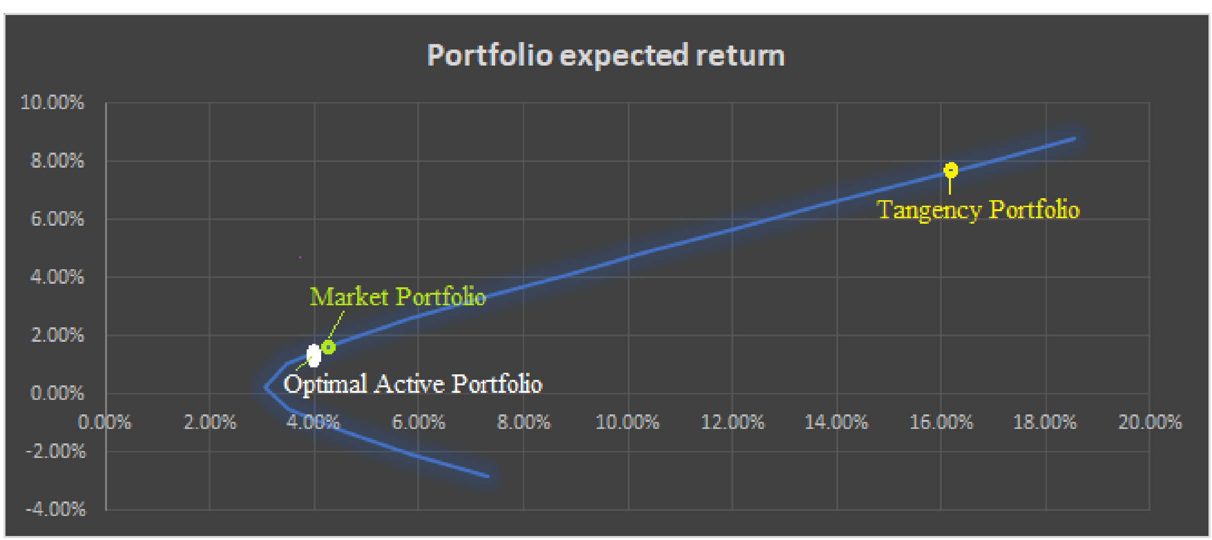


Figure.3: Efficient Frontier curve and the positions of the portfolios

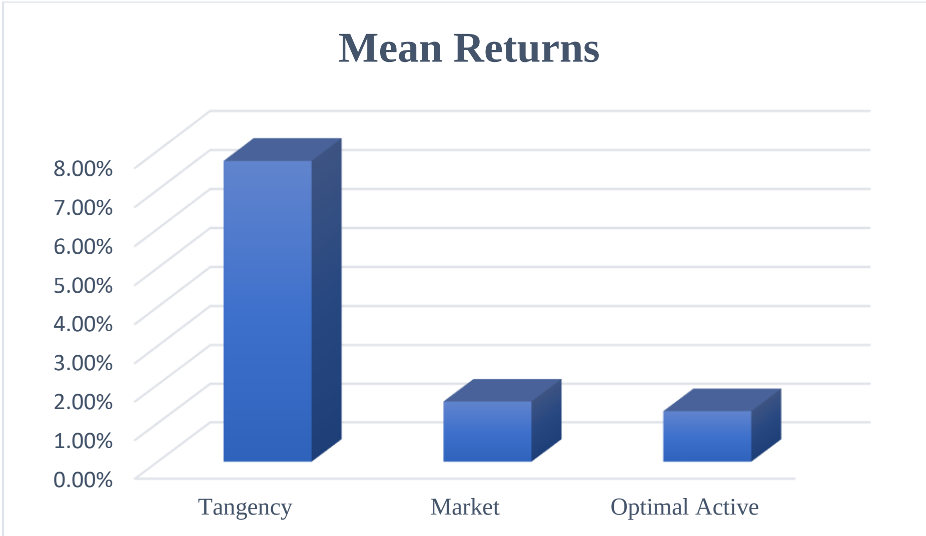


Figure.4: Mean returns chart.

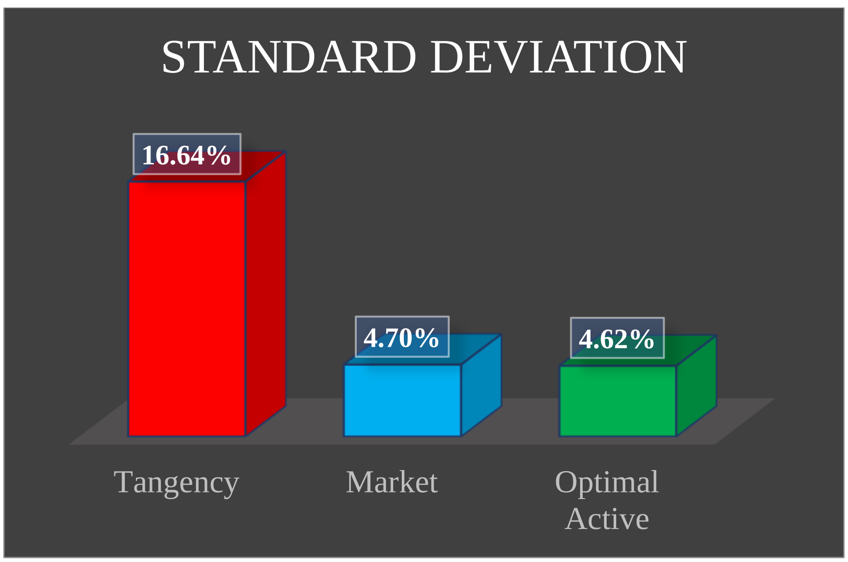


Figure.5: Standard deviation chart for the portfolios.

- Having high sharpe ratio [Figure.6] means the analyst's views are right, but here it is the least for the optimal active portfolio which may infer that the analyst's prediction of forecasts on outperformance of MRK and PFE could be wrong.

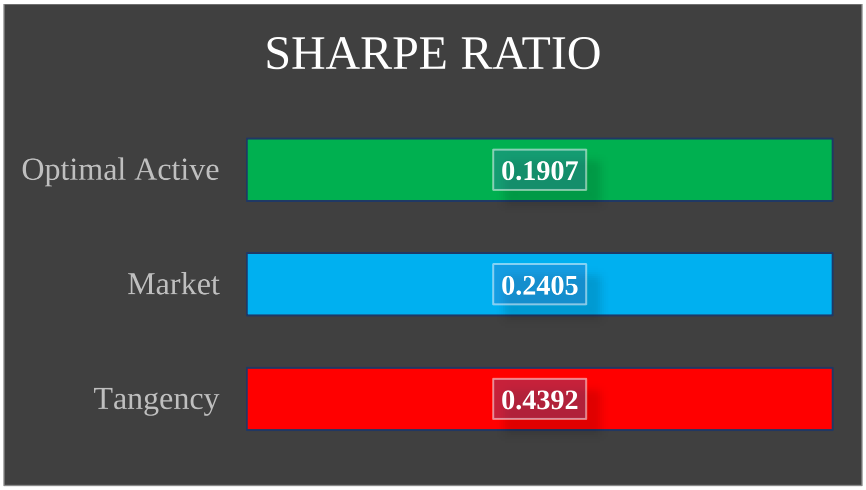


Figure.6: Sharpe ratio values

4.2.1-Month VaR and CVaR values:

With 99% confidence it can be estimated that the maximum loss in portfolio for anyone investing in the tangency portfolio for a monthly horizon would incur a maximum loss of 34.68% [Table 5], whilst the same parameter would result in a maximum loss of 9.3% for the market portfolio and 8.76% loss only for the Optimal Active Portfolio.

Table 5: Value at Risk and Conditional Value at Risk metrics for 99% confidence level for 1-month interval. The table also incorporates a sample risk prediction for an investor with a portfolio of $1Million.

Value at risk	Tangency Portfolio	Market Portfolio	Optimal Active Portfolio
99% 1-month VaR (zero mean)	34.68%	9.33%	8.76%
99% 1-month CVaR (zero mean)	39.96%	9.85%	9.12%
Sample Portfolio	$1,000,000.00	$1,000,000.00	$1,000,000.00
1-month $VaR	$346,827.73	$93,345.27	$87,604.54
1-month $CVaR	$399,550.29	$98,491.95	$91,247.69

The estimate of CVaR addresses the issues faced in VaR where it explains the average value of losses beyond the range of VaR, it is noticeable that the CVaR values are also directly proportional to that of the VaR values.

4.3.Sensitivity Analysis:

The analyst's views for MRK and PFE stocks are projected at 0.2% and 0.3%, respectively. However, these are estimated expectations. Based on the sensitivity analysis, if the analyst’s views for these assets fluctuate by ±0.2%, MRK's weight in the active portfolio could range from 19.94% to 36.25%, while PFE's weight could vary from 15.5% to 26.38%.

Table 6: Individual analyst's expectations vs corresponding portfolio weights.

MRK Analyst Expected views	Expected Active Portfolio weights	PFE Analyst Expected views	Expected Active Portfolio weights
0.10%	19.94%	0.20%	15.52%
0.12%	21.28%	0.22%	16.34%
0.14%	22.58%	0.24%	17.15%
0.16%	23.84%	0.26%	17.94%
0.18%	25.05%	0.28%	18.72%
0.20%	26.23%	0.30%	19.48%
0.22%	27.37%	0.32%	20.23%
0.24%	28.48%	0.34%	20.96%
0.26%	29.55%	0.36%	21.68%
0.28%	30.60%	0.38%	22.39%
0.30%	31.61%	0.40%	23.09%
0.32%	32.59%	0.42%	23.77%
0.34%	33.54%	0.44%	24.44%
0.36%	34.47%	0.46%	25.10%
0.38%	35.37%	0.48%	25.75%
0.40%	36.25%	0.50%	26.38%

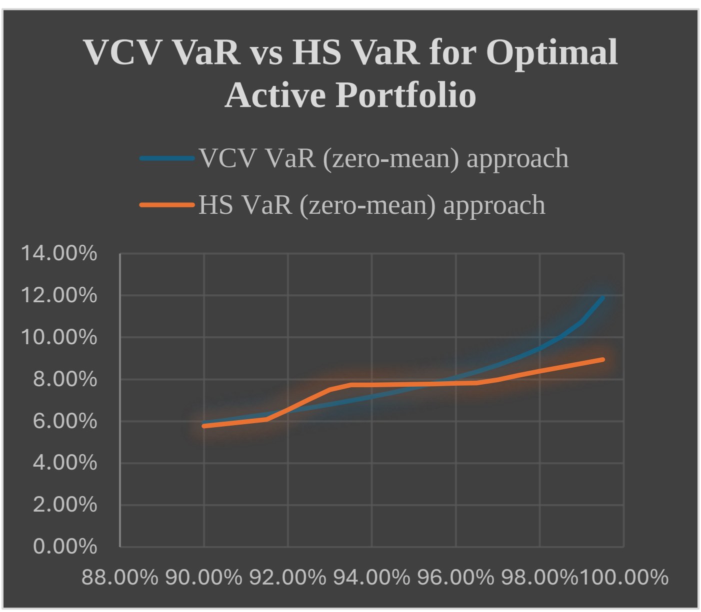
Table 7: Mean returns and Sharpe ratio sensitivity analysis

Tangency		Market		Optimal Active	
Mean ret	Sharpe ratio	Mean ret	Sharpe ratio	Mean ret	Sharpe ratio
1.50%	0.0642	1.50%	0.2274	1.50%	0.2314
2.50%	0.1243	2.50%	0.4404	2.50%	0.4481
3.50%	0.1844	3.50%	0.6534	3.50%	0.6647
4.50%	0.2444	4.50%	0.8663	4.50%	0.8814
5.50%	0.3045	5.50%	1.0793	5.50%	1.0981
6.50%	0.3646	6.50%	1.2923	6.50%	1.3147
7.50%	0.4247	7.50%	1.5052	7.50%	1.5314
8.50%	0.4848	8.50%	1.7182	8.50%	1.7481
9.50%	0.5449	9.50%	1.9312	9.50%	1.9648
10.50%	0.6050	10.50%	2.1441	10.50%	2.1814

- This chart [Table 7] was analyzed because the mean returns and Sharpe ratios for the market and active portfolios were lower, making them less comparable to the Tangency portfolio's higher mean return and Sharpe ratio values.
- Table 7 suggests that if the market and active portfolios achieved mean returns of 7.5%, their Sharpe ratios would exceed that of the Tangency portfolio, indicating better risk-adjusted performance.

Table 8: Variance Co-variance VaR vs Historical Simulation VaR for the portfolios

Sensitivity analysis of var-optimal active portfolio			Sensitivity analysis of var-tangency portfolio		
Confidence Interval	VCV VaR (0-mean)	HS VaR (0-mean)	Confidence Interval	VCV VaR (0-mean)	HS VaR(0-mean)
90.00%	5.91%	5.77%	90.00%	21.33%	21.93%
90.50%	6.05%	5.88%	90.50%	21.81%	22.12%
91.00%	6.19%	5.98%	91.00%	22.31%	22.31%
91.50%	6.33%	6.09%	91.50%	22.84%	22.50%
92.00%	6.48%	6.55%	92.00%	23.38%	23.35%
92.50%	6.64%	7.03%	92.50%	23.96%	24.24%
93.00%	6.81%	7.51%	93.00%	24.56%	25.13%
93.50%	6.99%	7.73%	93.50%	25.20%	25.67%
94.00%	7.18%	7.74%	94.00%	25.87%	25.93%
94.50%	7.38%	7.75%	94.50%	26.60%	26.19%
95.00%	7.59%	7.77%	95.00%	27.37%	26.50%
95.50%	7.82%	7.79%	95.50%	28.21%	27.06%
96.00%	8.08%	7.81%	96.00%	29.13%	27.61%
96.50%	8.36%	7.83%	96.50%	30.15%	28.16%
97.00%	8.68%	7.99%	97.00%	31.30%	28.91%
97.50%	9.05%	8.19%	97.50%	32.62%	29.72%
98.00%	9.48%	8.39%	98.00%	34.18%	30.53%
98.50%	10.02%	8.58%	98.50%	36.11%	32.05%
99.00%	10.74%	8.76%	99.00%	38.71%	34.68%
99.50%	11.89%	8.94%	99.50%	42.87%	37.32%



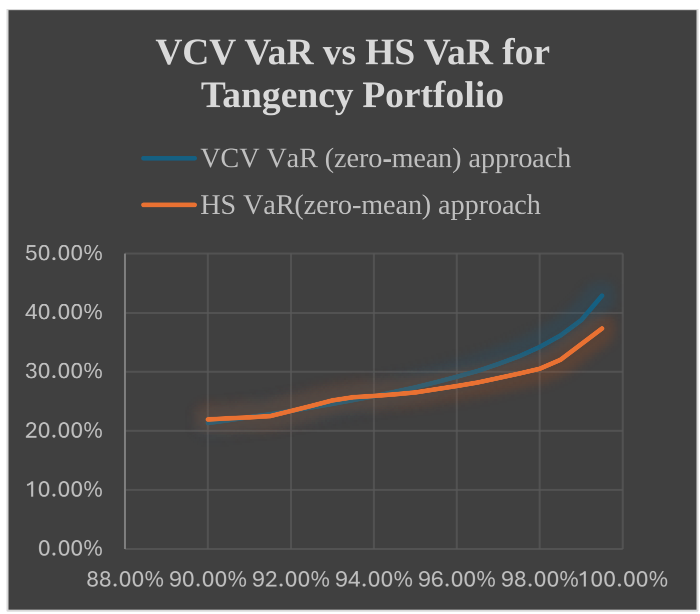


Figure.7: Comparison charts for both VaR approaches on Optimal active (Left) and Tangency portfolio (Right).

- The VCV VaR assumes a normal distribution of returns, which tends to overestimate risk, especially at higher confidence levels, leading to a steeper increase in VaR compared to HS VaR.

- HS VaR does not assume that returns are normally distributed. Instead, it uses actual historical returns data to calculate potential losses. If the historical period used does not contain extreme negative events, the HS VaR will reflect lower risk, as it does not predict extreme outcomes unless they have occurred in the past.
- Data-Driven HS VaR: The Historical Simulation VaR is based on actual past market data, which often reflects lower tail risk unless extreme events have occurred historically, resulting in a more gradual increase in VaR.
- Portfolio Sensitivity: The Tangency portfolio shows greater divergence between the methods at higher confidence levels, indicating its higher sensitivity to the assumptions underlying the VCV method.

5. Conclusion:

In conclusion, this analysis has applied financial modeling techniques to construct and assess portfolios within the US healthcare sector. Using the Markowitz mean-variance framework, the report effectively determined an efficient frontier, aiding in the selection of portfolios that achieve a balanced approach to risk and return. The Tangency Portfolio, Market Portfolio, and Optimal Active Portfolio were rigorously compared, offering valuable insights into their respective risk-return dynamics.

5.1. The main findings of this analysis:

- The Tangency Portfolio, while offering the highest expected return, also carries significantly higher risk, as evidenced by its superior Sharpe ratio and higher standard deviation. This portfolio is highly optimized for risk-return efficiency but may be unsuitable for investors with lower risk tolerance due to its concentration risk.
- Conversely, the Optimal Active Portfolio, constructed using the Black-Litterman approach, provides a more balanced strategy. It aligns closely with the market portfolio but still manages to enhance returns through strategic asset weighting, particularly in stocks like Merck and Eli Lilly.
- Due to very low Sharpe ratio of active portfolio with respect to tangency and market portfolio, the analyst's predictions of the outperformance of 2 stocks could be incorrect, the other potential alternative to this inference is either the calculations for estimation of portfolios is wrong or that the portfolio is modelled to receive the least sharpe ratio value.
- If the analysts's views are incorrect, then the sensitivity analysis of MRK and PFE with ±0.2% analyst's delta views indicate the range of possible set of weights for active portfolio.
- A what if analysis was performed on a hypothetical scenario to look at the risk adjusted return when market and optimal active portfolio had mean returns of 7.5%.
- An understanding of why VCV VaR had higher estimates in comparison to Historical Simulation VaR.
- Moreover, the assumption of normal distribution in the Variance-Covariance VaR approach could lead to an underestimation or overestimation of risk, particularly in extreme market conditions.
- The Historical Simulation VaR, while addressing some of these issues, still depends on past data, which may not always predict future market behavior accurately.

5.2. Limitations of this methodology:

- One critical shortcoming lies in the data itself. The analysis was based on a relatively short historical period of five years, which may not fully capture the long-term performance and risks associated with the healthcare sector.

- The use of monthly data points, while practical, may overlook intra-month volatility that could impact portfolio performance.
- The methodology, while robust, also relies heavily on historical data, which may not fully account for future market conditions, particularly in a sector as dynamic as healthcare.

5.3. Solutions to overcome the limitations:

- It would be beneficial to extend the data period to include more years, providing a more comprehensive view of long-term trends.
- Additionally, incorporating more frequent data points, such as daily returns, could offer a better understanding of volatility and risk.
- Exploring alternative risk measures, such as Stress Testing and Scenario Analysis, could also provide deeper insights into potential future risks (Fairhurst, 2015).
- Finally, integrating machine learning techniques to predict future returns and volatility could further enhance the precision and reliability of portfolio optimization in the healthcare sector.

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